HOW TO ...

... Upgrade Magneto X to:

- **Raspberry Pi from scratch**
- **Use Kalico instead of Klipper**
- **Use Nitehawk36 (USB) instead of original CAN-Toolboard**
- **Use BTT ebb36 because USB made too many problems**
- **Finally change to Fysetc H36 because more ports and higher temperature resistance**
- **Use BTT eddy instead of load cell (load cell is no more working after upgrade)**
- **Change eddy to beacon3D, because of not fully compatible eddy**
- **Implementation of Spoolman (I run it on a different Pi)**
- **Upgraded to Raspberry Pi 5 with PCIe to USB+NVMe HAT (no more SD card)**
- **CAUTION: I have some other modification on my machine like 0.9° motors, different lead screw, BoxTurtle, so my config is modded and I don't offer it here**
- **Requirement: You need a working magneto and a good configs before upgrading**

! USE AT YOUR OWN RISK !

! MACHINE'S WARRANTY MAY BE VOID !

! DOING IT WRONG CAN DAMAGE YOUR PRINTER !

! I AM NOT RESPONSIBLE FOR ANYTHING YOU DO !

>> I slow down to CAN speed to 250000 with 512 txqueue length. Looks like it is more reliabel on my machine <<

Copy and safe your config!
/home/pi/printer_data/config

Copy and safe Magneto specific files:
/home/pi/klipper/klippy/extras/gcode_shell_command.py
home/pi/klipper/klippy/extras/magneto_load_cell.py
/home/pi/auto-uuid
(can be found as well in github repo: https://github.com/mypeopoly/Klipper/tree/master)
Step 1: Install a Raspberry Pi image
use Raspberry Pi imager
Select Raspberry Pi OS other --> Raspberry Pi OS lite (64-bit or 32-bit)
Set network and SSH using "pi" and "raspberry"

IMPORTANT

BTT SKSM is NOT working with Raspberry Pi 5 (too much power needed for Pi 5)

It is necessary to use a high power (5 Amps) power device! The original Raspberry power supply is a little weak if you want to run several USB devices. There is a 45W power supply available which is better.
Use a separate power supply for each HAT

The Pi 5 should boot directly from NVMe without SD card, so you can use Pi Imager to write image directly to your NVMe and install it to the Raspberry hat.

After finishing the steps and your Pi is booting just fine we continue.

1st of all you need to do a complete upgrade/update:

sudo rpi-update
sudo apt update
sudo apt full-upgrade
sudo rpi-eeprom-update

if update is available:

sudo rpi-eeprom-update -d -a

restart with

sudo reboot

sudo apt-get update --fix-missing
sudo apt install rpi-update
sudo apt-get install -y crudini
sudo apt autoremove

restart with

sudo reboot

Then install git with

sudo apt-get update && sudo apt-get install git -y

Info: It is better to flash the new bootloader via command than via raspi-config.

Install the necessary packages

sudo apt install python3
python3 -m pip cache purge
~/klippy-env/bin/pip install -r ~/klipper/scripts/klippy-requirements.txt
sudo apt install python3-pip
sudo apt install python3-pil
sudo apt install python3-numpy
sudo apt install python3-flask
sudo apt install python3-serial
sudo apt install python3-scipy
python3 -m pip config set global.break-system-packages true
sudo apt install gfortran
sudo apt install cmake
sudo apt install libopenblas-dev
sudo pip3 install Rpi.GPIO --break-system-packages
sudo pip3 install spidev --break-system-packages
sudo pip3 install python-can --break-system-packages
python3 -m pip cache purge

Change boot order if using NVMe

sudo -E rpi-eeprom-config --edit
change boot order to:

BOOT_ORDER=0xf146

If not already add:

PCIE_PROBE=1

Save and reboot.

INFO: You can change boot order via raspi-config as well, but it will maybe not add the PCIE_PROBE=1. Some NVMe will not start properly without this order.

For Raspberry Pi 5 only

You can switch on/off the USB Power control via “sudo raspi-config”

4 Performance options --> USB Current --> switch off current limit

You can set PCIe to version 3 via “sudo raspi-config”

6 Advanced Options --> A8 PCIe speed

Cooling fan speed:

If you like to change the cooling fan speed you need to change the boot/firmware/config.txt

Add e.g. the following:

[all]
dtparam=fan_temp0=30000
dtparam=fan_temp0_hyst=3000
dtparam=fan_temp0_speed=65

dtparam=fan_temp1=55000
dtparam=fan_temp1_hyst=5000
dtparam=fan_temp1_speed=128

dtparam=fan_temp2=75000
dtparam=fan_temp2_hyst=5000
dtparam=fan_temp2_speed=154

dtparam=fan_temp3=80000
dtparam=fan_temp3_hyst=4000
dtparam=fan_temp3_speed=192

dtparam=fan_temp4=90000
dtparam=fan_temp4_hyst=4000
dtparam=fan_temp4_speed=255

 # add this for enabling maximum current for usb
usb_max_current_enable=1
#add this for PCIe in V3 mode
dtparam=pciex1_gen=3
 # add this to disable Bluetooth
dtoverlay=disable-bt

To remove the bluetooth package completely:
sudo systemctl disable hciuart.service
sudo systemctl disable bluealsa.service
sudo systemctl disable bluetooth.service

Disable WIFI power saver
check status with:
sudo iwconfig wlan0

Disable power saver with
sudo crontab -e

which opens the root crontab in your chosen editor... (or select your editor, usually nano)
add the following line at the bottom of the root crontab:

@reboot /usr/sbin/iw wlan0 set power_save off

Save, exit and reboot. Check again with *sudo iwconfig wlan0* if it is disabled.

```
pi@raspberrypi:~ $ sudo crontab -e
no crontab for root - using an empty one

Select an editor.  To change later, run 'select-editor'.
  1. /bin/nano        <---- easiest
  2. /usr/bin/vim.tiny
  3. /bin/ed

Choose 1-3 [1]:
crontab: installing new crontab
pi@raspberrypi:~ $ sudo iwconfig wlan0
wlan0     IEEE 802.11  ESSID:"3D-Druck"
          Mode:Managed  Frequency:5.32 GHz  Access Point: 88:FC:A6:23:8F:4D
          Bit Rate=433.3 Mb/s   Tx-Power=31 dBm
          Retry short limit:7   RTS thr:off   Fragment thr:off
          Encryption key:off
          Power Management:on
          Link Quality=70/70  Signal level=-34 dBm
          Rx invalid nwid:0  Rx invalid crypt:0  Rx invalid frag:0
          Tx excessive retries:3  Invalid misc:0   Missed beacon:0

pi@raspberrypi:~ $ sudo reboot

Broadcast message from root@raspberrypi on pts/2 (Mon 2025-05-26 20:22:01 CEST):

The system will reboot now!

pi@raspberrypi:~ $
Last login: Mon May 26 20:18:27 2025 from 192.168.178.74

SSH is enabled and the default password for the 'pi' user has not been changed.
This is a security risk - please login as the 'pi' user and type 'passwd' to set a new password.

pi@raspberrypi:~ $ sudo iwconfig wlan0
wlan0     IEEE 802.11  ESSID:"3D-Druck"
          Mode:Managed  Frequency:5.32 GHz  Access Point: 88:FC:A6:23:8F:4D
          Bit Rate=433.3 Mb/s   Tx-Power=31 dBm
          Retry short limit:7   RTS thr:off   Fragment thr:off
          Encryption key:off
          Power Management:off
          Link Quality=70/70  Signal level=-34 dBm
          Rx invalid nwid:0  Rx invalid crypt:0  Rx invalid frag:0
          Tx excessive retries:0  Invalid misc:0   Missed beacon:0

pi@raspberrypi:~ $
```

Steps to install Klipper / Kalico

Install KIAUH (https://github.com/dw-0/kiauh)

Use V6 !!!

Change repository to kalico (see https://github.com/KalicoCrew/kalico), `bleeding-edge-v2` as the new branch name

Install klipper, mainsail, klipperscreen, crowsnest via KIAUH

For Raspberry Pi5 and Klipperscreen it is neccessary to uise WAYLAND and NETWORK MANAGER

To have a faster reboot just disable waiting for network manager with:

sudo systemctl disable NetworkManager-wait-online.service

sudo systemctl mask NetworkManager-wait-online.service

--> USING A CAMERA WITH CROWSNEST AND TIMELAPSE CAN CAUSE USB FAILURES (LOST COMMUNICATION) IF USING A PI 5 <--

Install timelapse
https://github.com/mainsail-crew/moonraker-timelapse
Install your personal add ons (TMC autotune, Sonar, shaketune, ...):
https://github.com/ArmoredTurtle/AFC-Klipper-Add-On
https://github.com/julianschill/klipper-led_effect
https://github.com/andrewmcgr/klipper_tmc_autotune
https://github.com/mainsail-crew/sonar
https://github.com/Frix-x/klippain-shaketune
git clone https://github.com/mitant/peopoly-magneto-x-theme ~/printer_data/config/.theme
https://github.com/ArmoredTurtle/AFC-Klipper-Screen-Add-On
https://klipperbackup.xyz/
https://docs.beacon3d.com/quickstart/
https://github.com/moggieuk/KlipperScreen-Happy-Hare-Edition?tab=readme-ov-file
https://github.com/moggieuk/Happy-Hare/wiki/Installation

For Boxturtle only:
Update mainsail to **AFC** mainsail, just copy/paste the mainsail folder
(my copy is safed at: E:\3D-Drucker\config\Box Turtle\mainsail)

Steps to install magneto service and other files needed
Copy back to Raspberry Pi:
~~/home/pi/klipper/klippy/extras/gcode_shell_command.py~~ --> not necessary,
Kalico have its own

Copy back:
home/pi/klipper/klippy/extras/magneto_load_cell.py
/home/pi/auto-uuid
home/pi/printer_data/config

Create the magneto service file
sudo nano /etc/systemd/system/magneto.service
Copy and paste the text below into the nano editor

[Unit]
Description=Magneto Manager Service
After=network.target

[Service]
Type=simple
User=root
ExecStart=/bin/bash /home/pi/auto-uuid/magneto-run.sh

[Install]
WantedBy=multi-user.target

Change file permissions
Now, we need to change the file permissions to make it readable by all by typing
sudo chmod 644 /etc/systemd/system/magneto.service
As the last step, you need to tell the system that you have added this file and want to enable this service so that it starts at boot.
sudo systemctl daemon-reload
sudo systemctl enable magneto.service

change can config (if using CAN):
sudo nano /etc/network/interfaces.d/can0

----paste this:
allow-hotplug can0
iface can0 can static
    bitrate 250000
    up ifconfig $IFACE txqueuelen 512
    pre-up ip link set can0 type can bitrate 250000
    pre-up ip link set can0 txqueuelen 512

txqueuelen 1024 can work, but Kevin recommends to use txqueuelen 128, I set it to 512.

see also https://www.klipper3d.org/CANBUS.html

If you don't use the Magneto Linux-CAN-board you can use the standard setting:

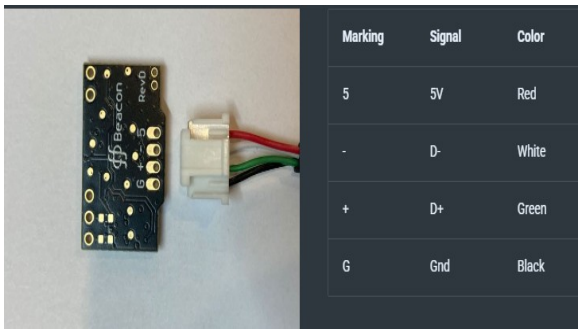
allow-hotplug can0
iface can0 can static
    bitrate 1000000
    up ifconfig $IFACE txqueuelen 128
    pre-up ip link set can0 type can bitrate 1000000
    pre-up ip link set can0 txqueuelen 128

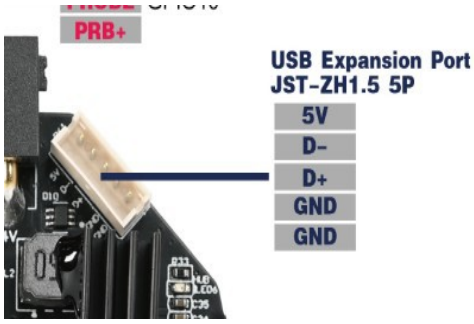
Install Beacon software

cd ~
git clone https://github.com/beacon3d/beacon_klipper.git
./beacon_klipper/install.sh **To install Beacon**

Strictly follow the guide on https://docs.beacon3d.com/quickstart/
Especialy don't forget to check your [stepper_z], it is necessary to have:
homing_retract_dist: 0

Wiring Beacon to nitehawk36:



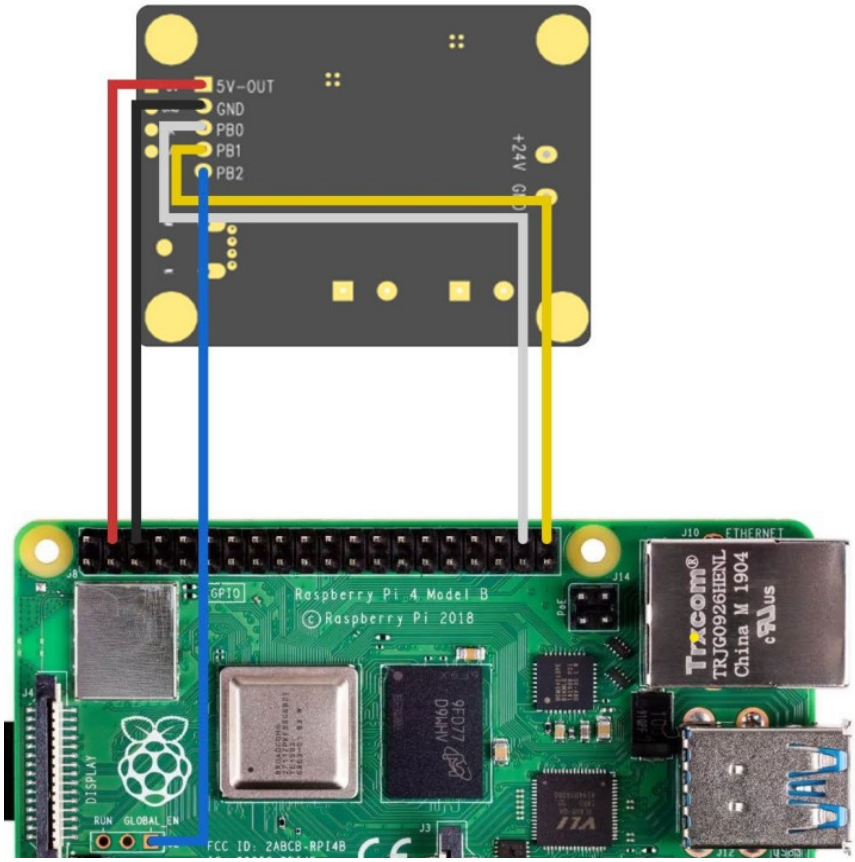


Helpful for adjustment:

Here the probe touches the bed and measures the distance. Then do the auto-calibrate. No more paper test necessary :-)
Example:
BEACON_AUTO_CALIBRATE SPEED=3.0 ACCEL=400.0 RETRACT=5.5 RETRACT_SPEED=3.0 SAMPLES=5 SAMPLES_TOLERANCE=0.0090 SAMPLES_TOLERANCE_RETRIES=5

For using BTT SKSM:

https://github.com/bigtreetech/SKSM
Insert the Micro SD card with the OS image into the computer. Add the following line to the config.txt file: dtoverlay=gpio-shutdown,gpio_pin=21. Then save and exit.
Wiring PB2 is not necessary



<u>IMPORTANT</u>

BTT SKSM is NOT working with Raspberry Pi 5 (too much power needed for Pi 5)

BTT Octopus firmware (necessary to flash for Kalico):

→ make menuconfig:

[*] Enable extra low-level configuration options

STMicroelectronics

STM32H723

128 kiB Bootloader

25 MHz crystal clock

communication interface (USB (on PA11/PA12))

(x) high precision stepping support

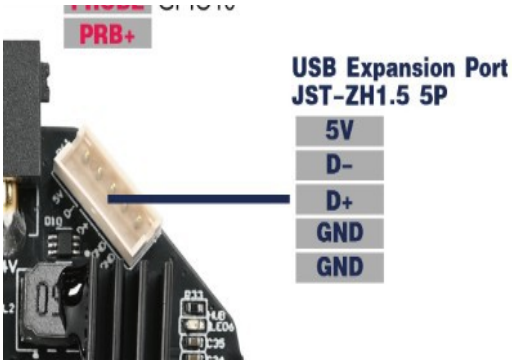
```
(Top)

[*] Enable extra low-level configuration options
    Micro-controller Architecture (STMicroelectronics STM32)  --->
    Processor model (STM32H723)  --->
    Bootloader offset (128KiB bootloader)  --->
    Clock Reference (25 MHz crystal)  --->
    Communication interface (USB (on PA11/PA12))  --->
    USB ids  --->
()  GPIO pins to set at micro-controller startup
[*] High-precision stepping support (slow)
```

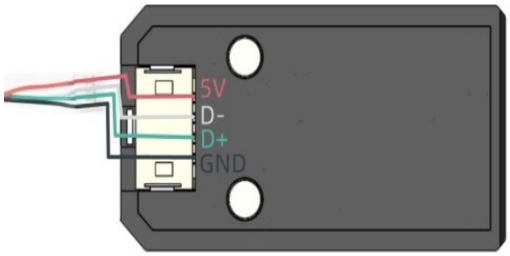
→ safe and close
→make clean
→make
Firmware is in output directory ~/klipper/out.

Flash Mainboard

*ls /dev/serial/by-id/**
*ls /dev/serial/by-path/**
sudo service klipper stop
make flash FLASH_DEVICE=/dev/serial/by-id/usb-Klipper_stm32h723xx_21003C001851333031373837-if00
udo service klipper start

Install BTT eddy and flash the device according to:
https://github.com/bigtreetech/Eddy
Wiring USB connection from nitehawk36 to eddy USB:
I used the original eddy cable, cut it and soldered it to the nitehawk36 USB extension cable.





Check and update all mcu serial with either
ls /dev/serial/by-id/*
or

ls /dev/serial/by-path/*
(by path can help to avoid timeout / lost communications)

Easiest way for me is to plug in one after the other USB device and run ls /dev/... after each plug to find out which is which.

Config hotend temperature

Make sure the <pullup_resistor> in [extruder] section is set to 2200 (nitehawk36 and H36)

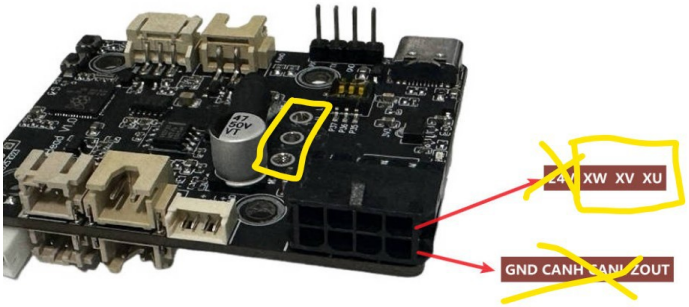
```
[extruder]
step_pin: nhk:gpio23
dir_pin: nhk:gpio24
enable_pin: !nhk:gpio25
rotation_distance: 4.637
microsteps: 16
full_steps_per_rotation: 200
nozzle_diameter: 0.400
filament_diameter: 1.75
heater_pin: nhk:gpio9
sensor_pin: nhk:gpio29
sensor_type: magneto_adc
max_extrude_only_distance: 200.0
max_extrude_cross_section: 45.0
pullup_resistor: 2200 #4700
max_temp: 310
max_power: 0.80
min_extrude_temp: 160
pressure_advance: 0.02
pressure_advance_smooth_time: 0.040
#control: pid
#pid_kp: 18.737 #11.745 #8.961
#pid_ki: 1.081 #0.535 #0.286
#pid_kd: 81.161 #64.473 #.276
```

Make sure the <pullup_resistor> in [extruder] section is set to 4700 (ebb36 only)

Magneto toolhead board

Now we can take out the toolhead board as well. We only need the U, V, W to be connected to the mover board. I soldered the wires directly to the top of the mover board and make a new plug for easier remove. Now 2 cables can be removed from the printer.

Take care for the wiring and double check it.



IMPORTANT

After doing this mod it is necessary to re-run the Linear motor host software to re-adjust the controller settings!!!!!

Filament sensor

nitehwak36	Magneto runout sensor
RGB Port 5V	VCC (blue)
RGB Port GND	GND (red)
Filament sensor +	Sensor (black)

Danger_Options (Kalico only)

Create a new file "danger_options.cfg"
Open it and copy/paste the danger options from
https://docs.kalico.gg/Config_Reference.html#danger-options

Read carefully through the options and set whatever you need, e.g. `#multi_mcu_trsync_timeout: 0.025` is maybe too small and need to set to e.g. 0.050.
Save
Open printer.cfg and include the new file
[include danger_options.cfg]

New PID routine

I am happy with my bed PID but not that much with extruder PID. It overshoots at about 8 to 10°C. Now with KALICO we can use another PID
change as follows

[extruder]
control = pid_v
pid_kp = 17.089

pid_ki = 1.238
pid_kd = 58.960

an re-calibrate with:
PID_CALIBRATE HEATER=extruder TARGET=250 WRITE_FILE=1 TOLERANCE=0.005

Extruder stepper motor (only nitehawk36)

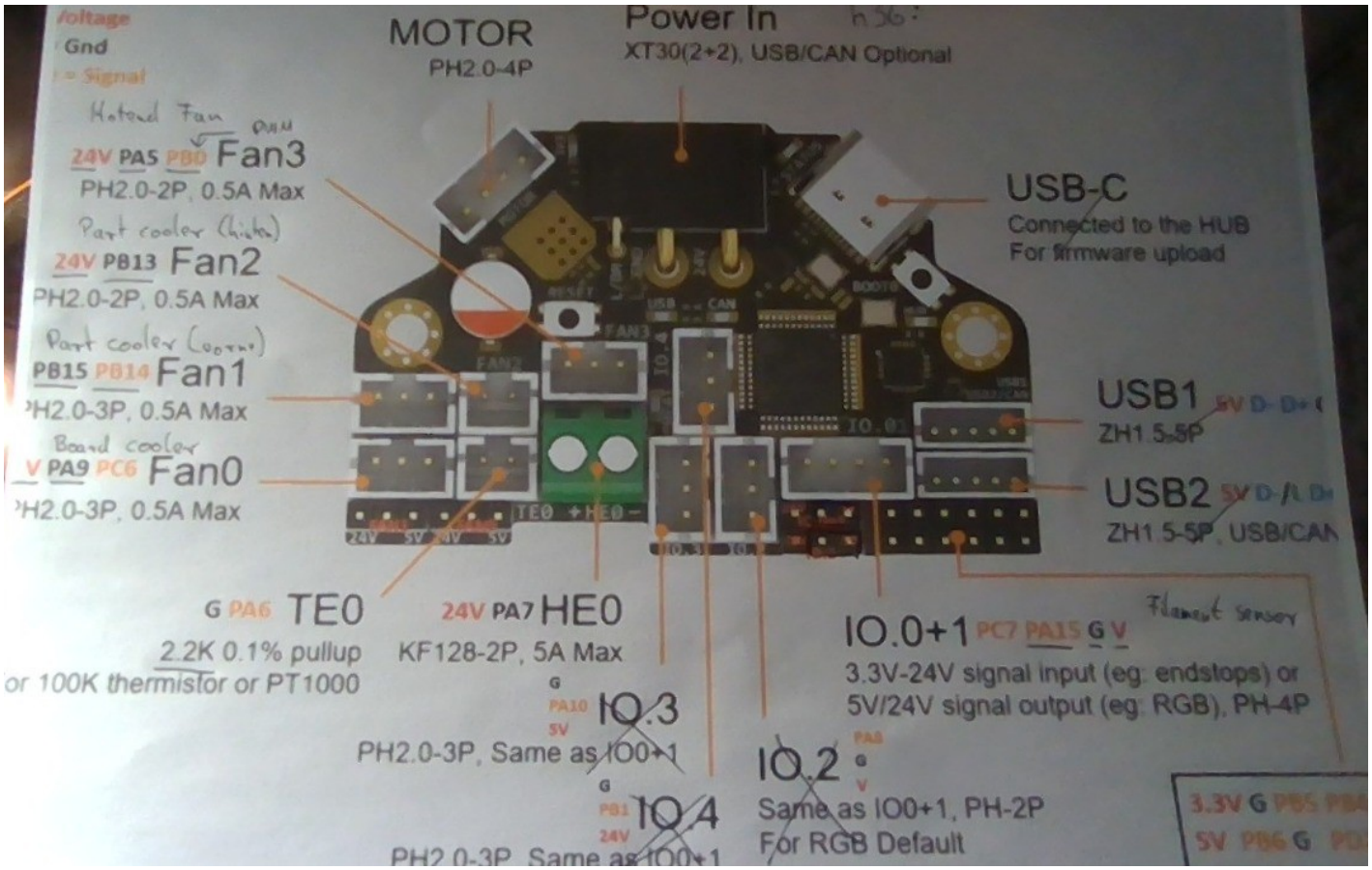
The nitehawk36 board delivers a sense_resistor of 0.100, so change it in your config

[tmc2209 extruder]
uart_pin: nhk:gpio0
interpolate: true
run_current: 1.0
#run and hold current are depending on motor you have.
#Keep it on the original values if you don't change the motor
#Additional cooling (cooling fan e.g.) is necessary with current higher than 0.7
hold_current: 0.6
sense_resistor: 0.100
stealthchop_threshold: 0

Additional cooling fan
The stands on the magneto toolhead can fit a 40 x 40 cooling fan. It can be wired to nitehawk36 Probe port, using GND and PRB+. Its always on, so a quite fan is recommended (Noctua e.g.). Voltage can be set to 24 or 5V by jumper.

Fysetc H36 Combo toolhead board

Pinout:



ATTENTION: This board switch the -24V, not the +24V!!! All fans will have +24V all time!!!

Find more information at https://wiki.fysetc.com/docs/H36-Combo

Firmware settings

```
(Top)
                        Klipper Firmware Configuration
[*] Enable extra low-level configuration options
    Micro-controller Architecture (STMicroelectronics STM32)  --->
    Processor model (STM32G0B1)  --->
    Bootloader offset (No bootloader)  --->
    Clock Reference (12 MHz crystal)  --->
    Communication interface (CAN bus (on PD0/PD1))  --->
(1000000) CAN bus speed
(!PA2) GPIO pins to set at micro-controller startup

[Space/Enter] Toggle/enter      [?] Help            [/] Search
[Q] Quit (prompts for save)     [ESC] Leave menu
```

Connect ONLY USB to your Raspberry Pi

Power cycle the printer

press and hold BOOT0,

- then press the RESET button for one second,
- then release RESET,
- wait for 3 seconds,
- and release BOOT0.

Check if device is in DFU mode with *lsusb*

flash new firmaware with

make flash FLASH_DEVICE=0483:df11

Find the canbus UUID with

sudo service klipper stop

~/klippy-env/bin/python ~/klipper/scripts/canbus_query.py can0

sudo service klipper restart

Spoolman
Create an entry in moonraker.conf:

[spoolman]
server: http://192.168.xxx.xxx:7912
 # URL to the Spoolman instance. This parameter must be provided.
sync_rate: 60
 # The interval, in seconds, between sync requests with the
 # Spoolman server. The default is 5.
BOXTURTLE – Spoolman:
delete --- it works now via web-interface

Fault: Neopixel update did not succeed
Modify the *klipper/klippy/extras/neopixel.py* file and increase the BIT_MAX_TIME parameter to 0.000030. After any change to the source code you must run a *sudo service klipper restart* for that change to take effect.

UPDATE ESP32 board
"The ESP32 is responsible for the buttons and LEDs at the bottom of the frame. It turns the motors on/off, handles errors, and similar. " (EmperorArthur)

Download the *.bin File:
https://github.com/EmperorArthur/magneto_x_linear_motor_controller_firmware/releases

Update using the Peopoly instructions:
https://wiki.peopoly.net/magneto/magneto-x/how-to-update-linear-motor-controller-firmware

Installation of the Waveshare 2-CH_CAN_HAT+
!!! Attention, don't mix it with the very similar 2-CH_CAN_FD_HAT, you see the difference if there is one (hat+) or two (fd_hat) jumper for 3.3/5V voltage selection!!!
refer to: *https://www.waveshare.com/wiki/2-CH_CAN_HAT+*

edit boot/firmware config.txt :
dtparam=spi=on
dtoverlay=i2c0
dtoverlay=spi1-3cs
dtoverlay=mcp2515,spi1-1,oscillator=16000000,interrupt=22
dtoverlay=mcp2515,spi1-2,oscillator=16000000,interrupt=13

Sudo reboot your Pi

Check with *dmesg | grep spi1*
you should receive something like:

```
(env) pi@raspberrypi:~ $ dmesg |grep spi
[    2.566685] mcp251x spi1.2 can0: MCP2515 successfully initialized.
[    2.587384] mcp251x spi1.1 can1: MCP2515 successfully initialized.
(env) pi@raspberrypi:~ $
```

Just for information
→ IFCONFIG HAS BEEN DEPRECATED. THE REPLACEMENT IS “IP ADDRESS” ←

Add CAN information to network:
Go to etc\network
create a directory called *interfaces.d (sudo mkdir interfaces.d)*
open the new directory
make two files:
sudo nano can0

insert:

auto can0
allow-hotplug can0
iface can0 can static
    bitrate 250000
    up ip link set $IFACE txqueuelen 512

close and safe

next file is very similar:
sudo nano can1

insert:

```
auto can1
allow-hotplug can1
iface can1 can static
        bitrate 250000
        up ip link set $IFACE txqueuelen 512
```

reboot with *sudo reboot*

```
sudo systemctl enable systemd-networkd
sudo systemctl start systemd-networkd
sudo systemctl disable systemd-networkd-wait-online.service
```

next is one row:
echo -e 'SUBSYSTEM=="net", ACTION=="change|add", KERNEL=="can" ATTR{tx_queue_len}="512"' | sudo tee /etc/udev/rules.d/10-can.rules > /dev/null*

next is one row:
echo -e "[Match]\nName=can\n\n[CAN]\nBitRate=250000\nRestartSec=0.1s\n\n[Link]\nRequiredForOnline=no" | sudo tee /etc/systemd/network/25-can.network > /dev/null*

After another *sudo reboot* and with *ip a* you should see something like this: (ip a is a shortcut for ip address)

```
[...]
4: can0: <NOARP,UP,LOWER_UP,ECHO> mtu 16 qdisc pfifo_fast state UP group default qlen 512
   link/can
5: can1: <NOARP,UP,LOWER_UP,ECHO> mtu 16 qdisc pfifo_fast state UP group default qlen 512
   link/can
```

State UP means, its running :-)

CAN txqueuelen different value after reboot
If you see the txqueuelen as a different number than set try the following (group default qlen set it to different value):
check:
etc/udev/rules.d/10-can.rules
etc/udev/rules.d/10-can-queue.rules
Both should look like this:
SUBSYSTEM=="net", ACTION=="change|add", KERNEL=="can" ATTR{tx_queue_len}="512"*

After change do the usual *sudo reboot* and check again with *ip a*

If you want the Peopoly logo in your mainsail web interface use this:
https://github.com/mitant/peopoly-magneto-x-theme

BTT EBB36
Very similar to nitehawk but with CAN support.

[mcu ebb36]
canbus_uuid: 84a63291b9dd
canbus_interface: can0

Follow the BTT guide to update the firmware. And find the ebb UUID.

In my setup I use the second CAN for EBB36.
Wiring at EBB36 is very limited. Need to do a splitter for part cooling fan.
There is no rpm feedback for the fans.
DIR_PIN for extruder need to be turned

[extruder]
step_pin: ebb36:PD0
dir_pin: !ebb36:PD1
enable_pin: !ebb36:PD2

Filament runout sensor is wired to the ebb36 endstop port PB6

Filament sensor

nitehwak36	ebb36	Magneto runout sensor
RGB Port 5V	+5V	VCC (blue)
RGB Port GND	GND	GND (red)
Filament sensor +	PB6	Sensor (black)

How to find CAN UUID
Stop Klipper and query the corrssponding CANBUS (CAN0 / CAN1)

```
sudo systemctl stop klipper.service
~/klippy-env/bin/python ~/klipper/scripts/canbus_query.py can0
~/klippy-env/bin/python ~/klipper/scripts/canbus_query.py can1
sudo systemctl start klipper.service
```

It may be possible that you have to do a complete powercycle before changes are working. EBB36 seems to need a powercycle before new firmware takes part.

Find my Configs on GitHub (branch: backup):
https://github.com/Schmudus/My-Magneto-X/tree/backup

SPECIAL THANKS TO:

Jetguy
mikebcbc
Arsoth
Deathzo
Infowolfe
EmperorArthur

and all others on discord who figured out how things could be done

– - - - - - TO BE CONTINUED - - - - - - -
(Next will be Fysetc H36 v1.3 toolhead board)

November, 09th 2025
Schmudi

HISTORY:

Version 2.4	:	PI5 added
Version 2.4.1	:	Error handling added
Version 2.4.2	:	Moonraker theme for Peopoly logo added
Version 2.4.3	:	Neopixel fault added
Version 2.4.4	:	small corrections
Version 2.4.5	:	small corrections + add picture for Raspberry WiFi power saver + update ESP32 board
Version 2.4.6	:	added information about Raspberry 5 bootloader, boot from NVMe and upgrade/update Raspberry Pi 5
Version 2.4.7	:	added information about upgrade/update Raspberry Pi 5
Version 2.4.8	:	change order to make it easier to follow delete unneccessary entries add more installations which are necessary for additional hardware like beacon
Version 2.4.9	:	added information to install waveshare 2 channel CAN HAT+
Version 2.4.10	:	add starting network.d service and small spell corrections
Version 2.4.11	:	add Peopoly Logo in mainsail
Version 2.4.12	:	CAN txqueuelen different value after reboot
Version 2.5.0	:	Added BTT EBB36 and corrected CAN settings
Version 2.5.1	:	small corrections and link to klipper backup added
Version 2.5.2	:	eeprom update added
Version 2.5.3	:	Find UUID added, GitHub added
Version 2.5.4	:	nothing new, order changed and formatting adjusted
Version 2.5.5	:	add Fysetc H36 Combo toolhead board
Version 2.5.6	:	fixed typo
Version 2.5.7	:	fixed update order
Version 2.5.8	:	add instruction to flash mainboard
Version 2.5.9	:	add additional information for Klipperscreen. Add Happ Hare installation.